Block practice · Ciphers: Caesar, Pigpen and Rail Fence

Work through each question type in order. Each section has a worked-example retry and all further practice. Attempt only the part outlined in red.

Pigpen cipher

Caesar cipher

Rail Fence cipher

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Practice workbook · Pigpen cipher

Using a Pigpen grid to turn letters into symbols or symbols into letters, classifying it as substitution, and judging whether a grid is a valid Pigpen key.

5 question parts: 1 worked-example retry and 4 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T3-21

1. Question 4 · part b · 3 marks

2022 JUN · Paper 02

Worked-example retry · Attempt part b only.

4 Several encryption algorithms have been developed.

(a) Identify what is meant by **encryption**.

(1)

- ☒ **A** Conversion of ciphertext into plaintext
- ☒ **B** Conversion of plaintext into ciphertext
- ☒ **C** Conversion of code into data
- ☒ **D** Conversion of data into information

(b) **Figure 1** shows a Pigpen cipher grid.

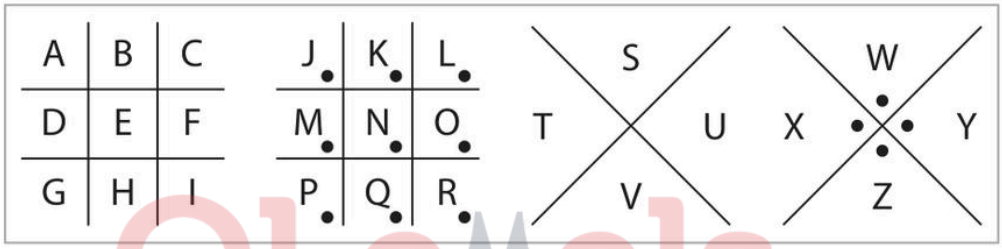


Figure 1

Complete the table by adding the symbol for each letter in the word **MAY**.

(3)

Letter	M	A	Y
Symbol			

(c) The message THE ENEMY IS NEAR is encoded using a different encryption algorithm.

Figure 2 shows the first stage of the encryption process.

T								I					
	H						Y		S				
		E				M				N			
			E		E						E		R
				N								A	

Figure 2

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

P 7 2 4 0 6 A 0 8 1 6

1. Question 4 · part b · 3 marks · Working and answer

Working and answer

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2. Question 4 · part a · 1 mark

2019 JUN · Paper 02

Further practice · Attempt part a only.

4 Encryption is used to encode data.

(a) Identify the word that describes a Pigpen cipher.

(1)

- ☒ **A** Plaintext
- ☒ **B** Random
- ☒ **C** Shift
- ☒ **D** Substitution

(b) **Figure 1** shows a Pigpen cipher grid.

(3)

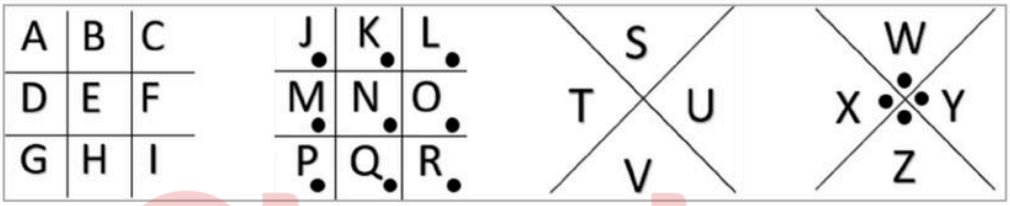


Figure 1

Complete the table to decode the symbols using the grid in **Figure 1**.

Symbol	Letter
	
	
	
	
	
	

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

10

P 6 1 8 8 1 R A 0 1 0 1 6

2. Question 4 · part a · 1 mark · Working and answer

Working and answer

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3. Question 4 · part b · 3 marks

2019 JUN · Paper 02

Further practice · Attempt part b only.

4 Encryption is used to encode data.

(a) Identify the word that describes a Pigpen cipher.

(1)

- ☒ **A** Plaintext
- ☒ **B** Random
- ☒ **C** Shift
- ☒ **D** Substitution

(b) **Figure 1** shows a Pigpen cipher grid.

(3)

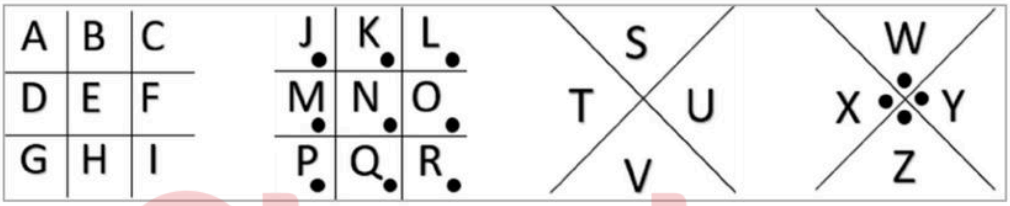


Figure 1

Complete the table to decode the symbols using the grid in **Figure 1**.

Symbol	Letter
	
	
	
	
	
	

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

10

P 6 1 8 8 1 R A 0 1 0 1 6

3. Question 4 · part b · 3 marks

2019 JUN · Paper 02

Further practice · Attempt part b only.

DO NOT WRITE IN THIS AREA

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DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(c) **Figure 2** shows another grid.

C	O	M	A	B	D	L	N	Q
P	U	T	F	G	H	V	W	X
E	R	S	I	J	K	Y	Z	

Figure 2

State whether this set of symbols meets the requirements of a Pigpen cipher.

Justify your answer.

(3)

State

Justification

(Total for Question 4 = 7 marks)

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3. Question 4 · part b · 3 marks · Working and answer

Working and answer

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4. Question 4 · part c · 3 marks

2019 JUN · Paper 02

Further practice · Attempt part c only.

4 Encryption is used to encode data.

(a) Identify the word that describes a Pigpen cipher.

(1)

- ☒ **A** Plaintext
- ☒ **B** Random
- ☒ **C** Shift
- ☒ **D** Substitution

(b) **Figure 1** shows a Pigpen cipher grid.

(3)

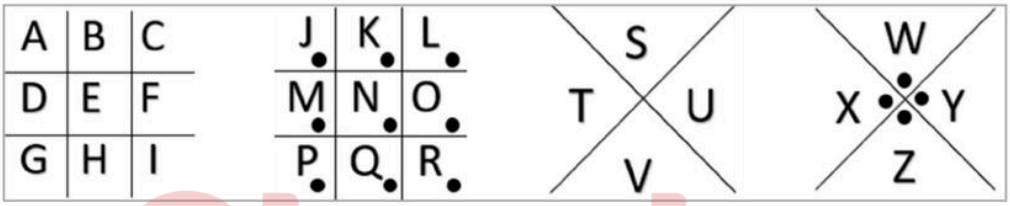


Figure 1

Complete the table to decode the symbols using the grid in **Figure 1**.

Symbol	Letter
	
	
	
	
	
	

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

4. Question 4 · part c · 3 marks

2019 JUN · Paper 02

Further practice · Attempt part c only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(c) **Figure 2** shows another grid.

C	O	M	A	B	D	L	N	Q
P	U	T	F	G	H	V	W	X
E	R	S	I	J	K	Y	Z	

Figure 2

State whether this set of symbols meets the requirements of a Pigpen cipher.

Justify your answer.

(3)

State

.....

Justification

.....

.....

.....

.....

(Total for Question 4 = 7 marks)

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4. Question 4 · part c · 3 marks · Working and answer

Working and answer

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5. Question 3 · part c · 2 marks

2025 JUN · Paper 01

Further practice · Attempt part c only.

DO NOT WRITE IN THIS AREA

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DO NOT WRITE IN THIS AREA

3 Computers use binary to represent and store data.

(a) Binary and hexadecimal can be used to represent numbers.

(i) Convert the binary pattern 1011 1001 to hexadecimal.

(2)

(ii) Identify the statement that is true for hexadecimal.

(1)

- ☒ **A** Hexadecimal can be processed faster by a computer than binary
- ☒ **B** Hexadecimal digits can be represented using one nibble
- ☒ **C** Hexadecimal uses less storage space in RAM than binary
- ☒ **D** Hexadecimal uses the number base 6

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(iii) Describe the binary shift that is equivalent to dividing an unsigned binary number by 4.

(2)

5. Question 3 · part c · 2 marks

2025 JUN · Paper 01

Further practice · Attempt part c only.

(b) **Figure 1** shows a bitmap image in a 16 × 16 grid. The image uses three colours.

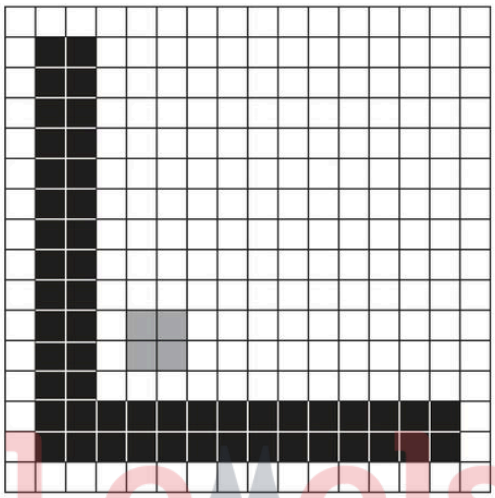


Figure 1

(i) State the smallest component of a bitmap image.

(1)

(ii) Construct an **expression** to show the number of images, like the one in **Figure 1**, that can be stored in 1 KB.

You do **not** need to do the calculation.

(4)

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5. Question 3 · part c · 2 marks

2025 JUN · Paper 01

Further practice · Attempt part c only.

(iii) Explain why a run-length encoding algorithm is a good choice for compressing the image in **Figure 1**.

(2)

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(c) The **Pigpen** cipher shown in **Figure 2** has been used to encrypt a message.

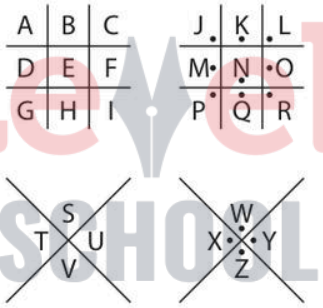
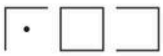


Figure 2

Complete the table to give the decrypted plain text.

(2)

Encrypted text	Plain text
	
	

(Total for Question 3 = 14 marks)

5. Question 3 · part c · 2 marks · Working and answer

Working and answer

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Practice workbook · Caesar cipher

Encrypting/decrypting with a Caesar shift, finding the shift, combining shifts, spotting wrap-around errors and explaining weakness.

4 question parts: 1 worked-example retry and 3 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T3-18

1. Question 4 · part a.i · 2 marks

2023 JUN · Paper 02

Worked-example retry · Attempt part a.i only.

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4 Data is encoded for encryption and for identifying records in databases.

(a) A Caesar cipher is one method of encryption.

(i) Complete the table to show the result of applying a Caesar cipher.

(2)

Plaintext	Shift	Ciphertext
PIXEL	−4	
CLOUD		FORXG

(ii) Plaintext is encrypted using a Caesar cipher.

Compare the ciphertexts produced for the plaintext **GOLD** by:

- a shift of −6 followed by a shift of +8
- a single shift of +2.

(2)

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(iii) Ann has written a Caesar cipher algorithm.

When the algorithm encrypts the plaintext **BYTE** with a shift of +5 the result is **GVYJ**.

This is incorrect.

Explain the error.

(2)

1. Question 4 · part a.i · 2 marks · Working and answer

Working and answer

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2. Question 3 · part b · 2 marks

2021 JUN · Paper 02

Further practice · Attempt part b only.

3 Several encryption algorithms have been developed.

(a) Ahmed has started to create a Caesar cipher encryption program.

Caesar cipher encryption works by giving a number value to a key.

Each letter in a plaintext message is replaced by a new ciphertext letter using the key.

For example, as shown in the table, a key value of +2 would change the plaintext message **encrypt** to the ciphertext message **gpetarv**.

Plaintext	e	n	c	r	y	p	t
Ciphertext	g	p	e	t	a	r	v

Ahmed's program already converts from plaintext to ciphertext.

He wants to extend the program to:

- accept the input of a plaintext message
- accept the input of the key, which must be a positive integer between 1 and 25
- write the generated ciphertext to a text file named **Cipher.txt**
- display the ciphertext.

Open **Q03a** in the code editor.

Complete the program code.

You must use the structure given in the file **Q03a** to complete the program.

Do not add any further functionality.

Save your code as **Q03aFINISHED** with the correct file extension for the programming language.

(8)

(b) Explain **one** reason why the Caesar cipher is a weak encryption algorithm.

(2)

.....

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DO NOT WRITE IN THIS AREA

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2. Question 3 · part b · 2 marks

2021 JUN · Paper 02

Further practice · Attempt part b only.

(c) The Vigenere cipher encrypts plaintext using a series of Caesar ciphers, based on the letters of a keyword and a Vigenere square.

Figure 1 shows a Vigenere square.

	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
a	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
b	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a
c	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b
d	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c
e	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d
f	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e
g	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f
h	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g
i	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h
j	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i
k	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j
l	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k
m	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l
n	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m
o	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n
p	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o
q	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p
r	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q
s	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r
t	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s
u	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t
v	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u
w	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v
x	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w
y	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x
z	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y

Figure 1

Complete the table to perform a Vigenere decryption of the ciphertext message **flmktrwhee** using the keyword **thirty**.

(4)

Encrypted letter	f	l	m	k	t	r	w	h	e	e
Keyword letter										
Decrypted letter										

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

P 6 6 4 9 0 A 0 9 1 6

9

Turn over ▶

2. Question 3 · part b · 2 marks · Working and answer

Working and answer

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3. Question 4 · part a.ii · 2 marks

2023 JUN · Paper 02

Further practice · Attempt part a.ii only.

DO NOT WRITE IN THIS AREA

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4 Data is encoded for encryption and for identifying records in databases.

(a) A Caesar cipher is one method of encryption.

(i) Complete the table to show the result of applying a Caesar cipher.

(2)

Plaintext	Shift	Ciphertext
PIXEL	−4	
CLOUD		FORXG

(ii) Plaintext is encrypted using a Caesar cipher.

Compare the ciphertexts produced for the plaintext **GOLD** by:

- a shift of −6 followed by a shift of +8
- a single shift of +2.

(2)

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(iii) Ann has written a Caesar cipher algorithm.

When the algorithm encrypts the plaintext **BYTE** with a shift of +5 the result is **GVYJ**.

This is incorrect.

Explain the error.

(2)

3. Question 4 · part a.ii · 2 marks · Working and answer

Working and answer

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4. Question 4 · part a.iii · 2 marks

2023 JUN · Paper 02

Further practice · Attempt part a.iii only.

4 Data is encoded for encryption and for identifying records in databases.

(a) A Caesar cipher is one method of encryption.

(i) Complete the table to show the result of applying a Caesar cipher.

(2)

Plaintext	Shift	Ciphertext
PIXEL	−4	
CLOUD		FORXG

(ii) Plaintext is encrypted using a Caesar cipher.

Compare the ciphertexts produced for the plaintext **GOLD** by:

- a shift of −6 followed by a shift of +8
- a single shift of +2.

(2)

OLevels.com SCHOOL

(iii) Ann has written a Caesar cipher algorithm.

When the algorithm encrypts the plaintext **BYTE** with a shift of +5 the result is **GVYJ**.

This is incorrect.

Explain the error.

(2)

DO NOT WRITE IN THIS AREA

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DO NOT WRITE IN THIS AREA

4. Question 4 · part a.iii · 2 marks

2023 JUN · Paper 02

Further practice · Attempt part a.iii only.

(b) A program is needed to create a key for a database.

A user enters a two-letter word, a whole number and a decimal number.

The program must ensure the word is only two characters long.

The program must display an error message when the word is not two characters long.

A key is generated from the whole number, the reversed word and the whole number part of the decimal number.

Figure 4 shows the input values and a valid key.

```
Enter a word: at
Enter a whole number: 12
Enter a decimal number: 7.89
12ta7
```

Figure 4

Open **Q04b** in the code editor.

Write the program.

Do not add any further functionality.

Save your code as **Q04bFINISHED** with the correct file extension for the programming language.

(8)

(Total for Question 4 = 14 marks)

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DO NOT WRITE IN THIS AREA

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4. Question 4 · part a.iii · 2 marks · Working and answer

Working and answer

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Practice workbook · Rail Fence cipher

Zig-zag transposition: writing text over n rails, reading off ciphertext, finding the key from a diagram, naming the method and explaining why it is weak.

5 question parts: 1 worked-example retry and 4 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T3-20

1. Question 3 · part c.i · 4 marks

2020 NOV · Paper 02

Worked-example retry · Attempt part c.i only.

3 Several encryption algorithms have been developed.

(a) State what is meant by the term 'encryption'.

(1)

(b) Give **one** reason why data may need to be encrypted.

(1)

(c) (i) Perform a Rail Fence cipher encryption of the text **COMPUTATIONAL** using a key of 4.

Show your working.

(4)

Encrypted text:

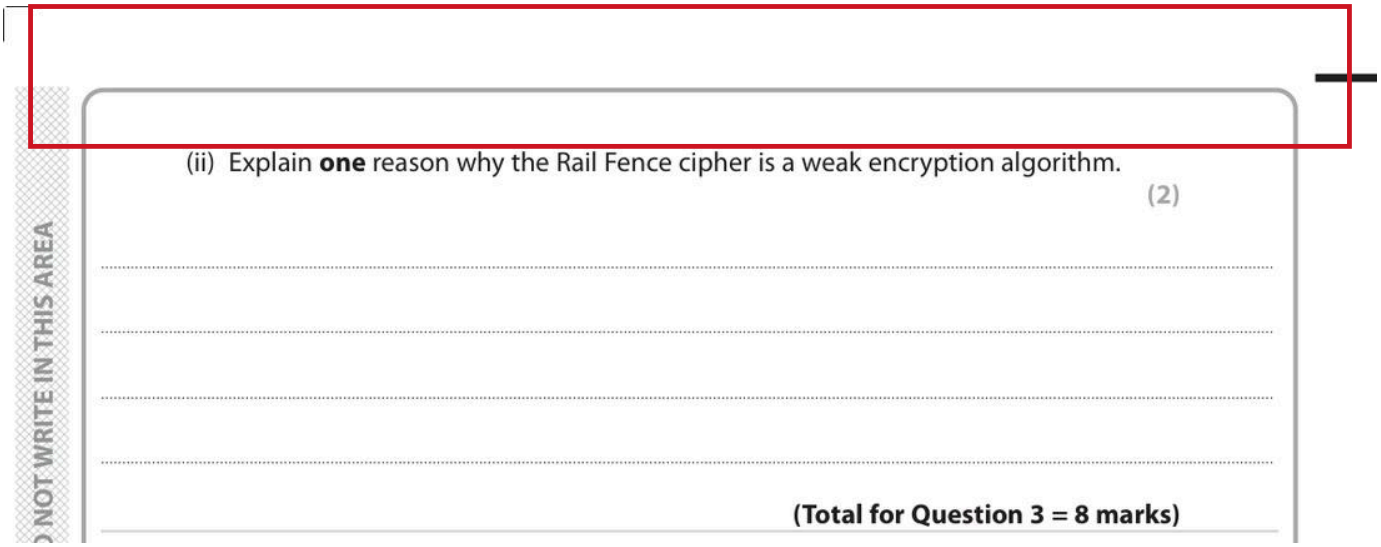
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

P 6 1 8 8 5 A 0 6 1 6

1. Question 3 · part c.i · 4 marks

2020 NOV · Paper 02

Worked-example retry · Attempt part c.i only.



DO NOT WRITE IN THIS AREA

(ii) Explain **one** reason why the Rail Fence cipher is a weak encryption algorithm.

(2)

(Total for Question 3 = 8 marks)

Working and answer

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2. Question 3 · part c.ii · 2 marks

2020 NOV · Paper 02

Further practice · Attempt part c.ii only.

3 Several encryption algorithms have been developed.

(a) State what is meant by the term 'encryption'.

(1)

(b) Give **one** reason why data may need to be encrypted.

(1)

(c) (i) Perform a Rail Fence cipher encryption of the text **COMPUTATIONAL** using a key of 4.

Show your working.

(4)

Encrypted text:

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

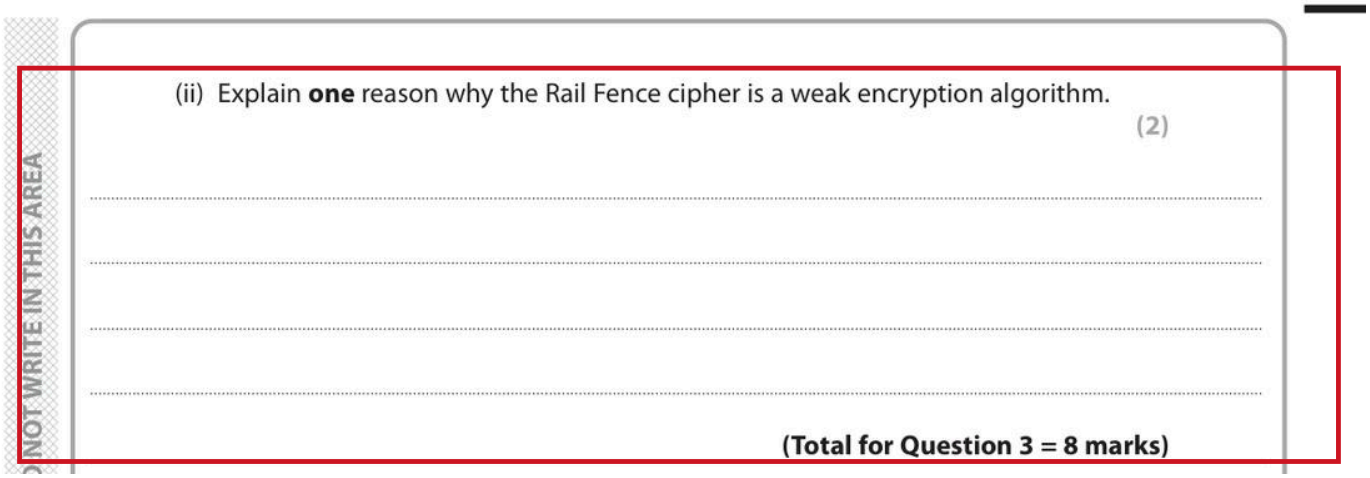
6

P 6 1 8 8 5 A 0 6 1 6

2. Question 3 · part c.ii · 2 marks

2020 NOV · Paper 02

Further practice · Attempt part c.ii only.


DO NOT WRITE IN THIS AREA

(ii) Explain **one** reason why the Rail Fence cipher is a weak encryption algorithm.

(2)

(Total for Question 3 = 8 marks)

Working and answer

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3. Question 4 · part c.i · 1 mark

2022 JUN · Paper 02

Further practice · Attempt part c.i only.

4 Several encryption algorithms have been developed.

(a) Identify what is meant by **encryption**.

(1)

- ☒ **A** Conversion of ciphertext into plaintext
- ☒ **B** Conversion of plaintext into ciphertext
- ☒ **C** Conversion of code into data
- ☒ **D** Conversion of data into information

(b) **Figure 1** shows a Pigpen cipher grid.

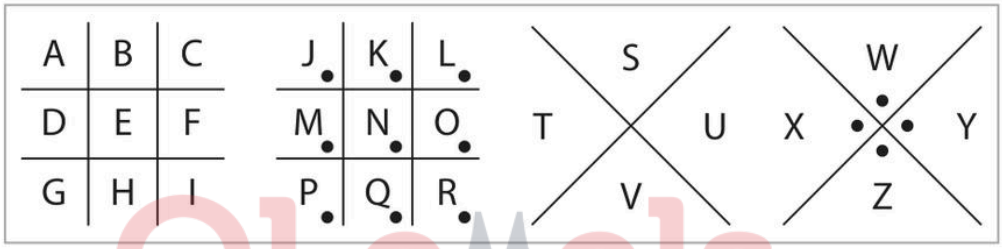


Figure 1

Complete the table by adding the symbol for each letter in the word **MAY**.

(3)

Letter	M	A	Y
Symbol			

(c) The message THE ENEMY IS NEAR is encoded using a different encryption algorithm.

Figure 2 shows the first stage of the encryption process.

T								I					
	H						Y		S				
		E				M				N			
			E		E						E		R
				N								A	

Figure 2

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

3. Question 4 · part c.i · 1 mark

2022 JUN · Paper 02

Further practice · Attempt part c.i only.

DO NOT WRITE IN THIS AREA

(i) Give the key used in the first stage of the encryption process.

(1)

(ii) Give the ciphertext that is produced by the encryption algorithm.

(1)

(iii) Give the name of the encryption algorithm that has been used to produce the ciphertext.

(1)

(Total for Question 4 = 7 marks)

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Working and answer

4. Question 4 · part c.ii · 1 mark

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Further practice · Attempt part c.ii only.

4 Several encryption algorithms have been developed.

(a) Identify what is meant by **encryption**.

(1)

- ☒ **A** Conversion of ciphertext into plaintext
- ☒ **B** Conversion of plaintext into ciphertext
- ☒ **C** Conversion of code into data
- ☒ **D** Conversion of data into information

(b) **Figure 1** shows a Pigpen cipher grid.

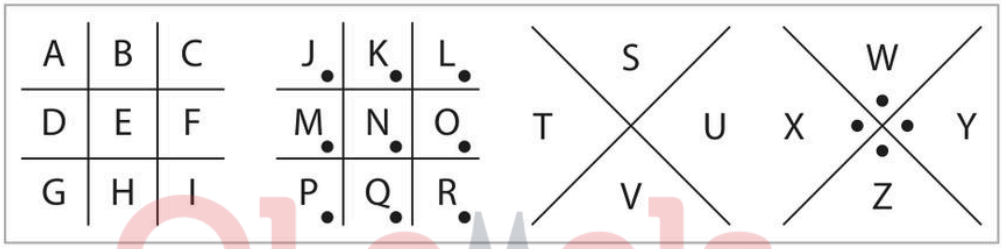


Figure 1

Complete the table by adding the symbol for each letter in the word **MAY**.

(3)

Letter	M	A	Y
Symbol			

(c) The message THE ENEMY IS NEAR is encoded using a different encryption algorithm.

Figure 2 shows the first stage of the encryption process.

T								I					
	H						Y		S				
		E				M				N			
			E		E						E		R
				N								A	

Figure 2

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

P 7 2 4 0 6 A 0 8 1 6

4. Question 4 · part c.ii · 1 mark

2022 JUN · Paper 02

Further practice · Attempt part c.ii only.

DO NOT WRITE IN THIS AREA

(i) Give the key used in the first stage of the encryption process.

(1)

(ii) Give the ciphertext that is produced by the encryption algorithm.

(1)

(iii) Give the name of the encryption algorithm that has been used to produce the ciphertext.

(1)

(Total for Question 4 = 7 marks)

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Working and answer

5. Question 4 · part c.iii · 1 mark

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Further practice · Attempt part c.iii only.

4 Several encryption algorithms have been developed.

(a) Identify what is meant by **encryption**.

(1)

- ☒ **A** Conversion of ciphertext into plaintext
- ☒ **B** Conversion of plaintext into ciphertext
- ☒ **C** Conversion of code into data
- ☒ **D** Conversion of data into information

(b) **Figure 1** shows a Pigpen cipher grid.

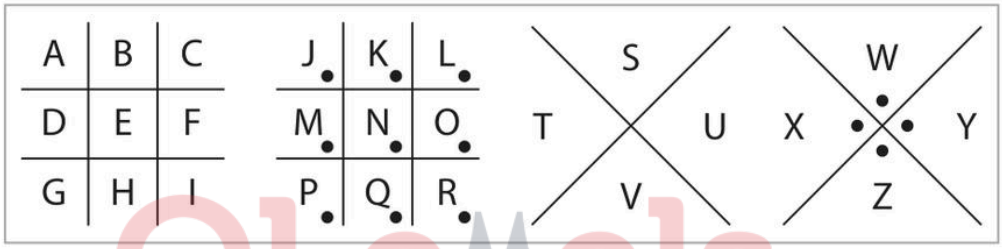


Figure 1

Complete the table by adding the symbol for each letter in the word **MAY**.

(3)

Letter	M	A	Y
Symbol			

(c) The message THE ENEMY IS NEAR is encoded using a different encryption algorithm.

Figure 2 shows the first stage of the encryption process.

T								I					
	H						Y		S				
		E				M				N			
			E		E						E		R
				N								A	

Figure 2

8

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

5. Question 4 · part c.iii · 1 mark

2022 JUN · Paper 02

Further practice · Attempt part c.iii only.

DO NOT WRITE IN THIS AREA

(i) Give the key used in the first stage of the encryption process.

(1)

(ii) Give the ciphertext that is produced by the encryption algorithm.

(1)

(iii) Give the name of the encryption algorithm that has been used to produce the ciphertext.

(1)

(Total for Question 4 = 7 marks)

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Working and answer